1. Which of the following is true about 1NF?

Answer: (c) All attributes must be atomic

Explanation:
First Normal Form (1NF) requires every attribute in a relation to contain atomic (single) values. A cell should not contain multiple or repeating values.

For example, if a student has two phone numbers, storing both numbers in one cell violates 1NF.

Incorrect:

Student_ID	Name	Phone
101	Ravi	9876543210, 9123456780

Correct 1NF representation:

Student_ID	Name	Phone
101	Ravi	9876543210
101	Ravi	9123456780

Thus, 1NF ensures atomic values and removes repeating groups. Partial dependencies are removed in 2NF, while transitive dependencies are removed in 3NF.

Hence, the correct answer is (c) All attributes must be atomic.

2. A relation is in 2NF if it is in 1NF and:

Answer: (b) Has no partial dependencies

Explanation:
Second Normal Form (2NF) requires a relation to first be in 1NF and then have no partial dependencies.

A partial dependency occurs when a non-prime attribute depends only on a part of a composite candidate key.

For example, consider:

ENROLLMENT(Student_ID, Course_ID, Student_Name, Course_Name, Grade)

If (Student_ID, Course_ID) is the candidate key:

- Student_ID → Student_Name
- Course_ID → Course_Name

- (Student_ID, Course_ID) → Grade

Student_Name and Course_Name depend only on part of the composite key. Therefore, partial dependencies exist.

The relation can be decomposed into:

- **STUDENT(Student_ID, Student_Name)**
- **COURSE(Course_ID, Course_Name)**
- **ENROLLMENT(Student_ID, Course_ID, Grade)**

Thus, partial dependencies are eliminated.

Hence, the correct answer is (b) Has no partial dependencies.

3. Which normal form deals with multi-valued dependencies?

Answer: (d) 4NF

Explanation:
Fourth Normal Form (4NF) deals with multivalued dependencies (MVDs).

A multivalued dependency occurs when one attribute determines multiple independent values of another attribute.

For example:

STUDENT(Student_ID, Hobby, Language)

A student may have multiple hobbies and multiple languages, and these two sets of values may be independent.

The dependencies can be represented as:

Student_ID →→ Hobby

Student_ID →→ Language

To remove the redundancy, the relation can be divided into:

- **STUDENT_HOBBY(Student_ID, Hobby)**
- **STUDENT_LANGUAGE(Student_ID, Language)**

Therefore, 4NF is specifically concerned with eliminating undesirable multivalued dependencies.

Hence, the correct answer is (d) 4NF.

4. BCNF is a stricter version of:

Answer: (c) 3NF

Explanation:
BCNF stands for Boyce-Codd Normal Form. It is a stricter form of Third Normal Form (3NF).

For a relation to be in BCNF, for every non-trivial functional dependency:

X → Y

X must be a superkey.

3NF has less strict requirements than BCNF. Therefore, a relation can sometimes be in 3NF but not in BCNF.

The normal-form progression is:

1NF → 2NF → 3NF → BCNF → 4NF → 5NF

BCNF helps remove additional redundancy caused by functional dependencies.

Hence, the correct answer is (c) 3NF.

5. Functional dependency X → Y means:

Answer: (b) X determines Y

Explanation:
A functional dependency is represented as:

X → Y

It means that the value of X uniquely determines the value of Y.

Here, X is called the determinant, while Y is the dependent attribute.

For example:

Student_ID → Student_Name

This means that knowing the Student_ID allows us to determine exactly one Student_Name.

For example:

Student_ID	**Student_Name**
101	Ravi
102	Priya
103	Arun

If Student_ID = 101, the student's name must be Ravi.

However, Student_Name → Student_ID does not necessarily hold because two students may have the same name.

Hence, the correct answer is (b) X determines Y.

6. Which of the following anomalies is NOT associated with unnormalized relations?

Answer: (d) Retrieval anomaly

Explanation:
Poorly designed or unnormalized relations can cause three major types of anomalies:

1. Insertion Anomaly:
Occurs when new information cannot be inserted without adding unrelated information.

2. Deletion Anomaly:
Occurs when deleting one record unintentionally removes other important information.

3. Update Anomaly:
Occurs when the same information is stored in multiple rows and must be updated at several places.

For example, if a department name is repeated for many employees, changing the department name requires updating many records. Failure to update all records causes an update anomaly.

These anomalies are reduced through database normalization.

Retrieval anomaly is not one of the standard three anomalies associated with unnormalized relations.

Hence, the correct answer is (d) Retrieval anomaly.

7. A superkey is:

Answer: (b) Any set of attributes that uniquely identifies tuples

Explanation:

A superkey is a set of one or more attributes that can uniquely identify each tuple in a relation.

Consider:

STUDENT(Student_ID, Name, Email, Department)

If Student_ID uniquely identifies every student, then the following are superkeys:

- {Student_ID}
- {Student_ID, Name}
- {Student_ID, Email}
- {Student_ID, Department}

A candidate key is a minimal superkey. For example, {Student_ID} is a candidate key because it contains no unnecessary attribute.

Therefore:

Superkey: Any set of attributes that uniquely identifies a tuple.

Candidate Key: A minimal superkey.

Primary Key: A candidate key selected to identify records.

Hence, the correct answer is (b) Any set of attributes that uniquely identifies tuples.

8. If R(A,B,C) has FD: A → B and B → C, then C is:

Answer: (c) Transitively dependent on A

Explanation:

Given:
$A \rightarrow B$
$B \rightarrow C$

using the transitivity property:

$A \rightarrow B \rightarrow C$

Therefore:

$A \rightarrow C$

C depends on A through the intermediate attribute B. This is called a transitive dependency.

For example:

Student_ID → Department_ID

and

Department_ID → Department_Name

Therefore:

Student_ID → Department_Name

Here, Department_Name is transitively dependent on Student_ID.

Transitive dependencies are removed during the conversion from 2NF to 3NF.

Hence, the correct answer is (c) Transitively dependent on A.

9. 5NF is also known as:

Answer: (c) Project-Join Normal Form

Explanation:
Fifth Normal Form (5NF) is also known as Project-Join Normal Form (PJ/NF).

5NF deals with join dependencies. It ensures that a relation cannot be further decomposed into smaller relations without losing information or creating undesirable combinations when the relations are joined.

For example, a relation involving:

Supplier, Part, and Project

may sometimes be decomposed into smaller relations such as:

- **SUPPLIER_PART**
- **SUPPLIER_PROJECT**
- **PART_PROJECT**

5NF ensures that such decomposition and subsequent joining does not introduce incorrect information.

Therefore:

5NF = Project-Join Normal Form (PJ/NF)

Hence, the correct answer is (c) Project-Join Normal Form.

10. Lossless-join decomposition ensures:

Answer: (a) No data is lost during decomposition

Explanation:
During normalization, a large relation is divided into smaller relations to reduce redundancy. This process is called decomposition.

A decomposition is called lossless-join if the original relation can be reconstructed exactly by joining the decomposed relations.

For example:

EMPLOYEE(Emp_ID, Emp_Name, Dept_ID, Dept_Name)

can be decomposed into:

EMPLOYEE(Emp_ID, Emp_Name, Dept_ID)

DEPARTMENT(Dept_ID, Dept_Name)

The two relations can be joined using Dept_ID to reconstruct the original information.

A lossless decomposition ensures:

- No valid information is lost.
- The original relation can be recovered.
- No incorrect tuples are generated.

Lossless join is an important property of normalization because reducing redundancy should not result in loss of information.

Hence, the correct answer is (a) No data is lost during decomposition.